Shushil Karki | Software Engineer

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PROFESSIONAL SUMMARY:

Full Stack Developer with 1.5+ years of experience architecting high-performance financial platforms and microservices. Expert in building scalable backends with Java (Spring Boot/MVC) and Node.js (NestJS/Express), integrated with dynamic React/TypeScript frontends using Next.js and Redux. Specialist in engineering event-driven architectures via Kafka/RabbitMQ and managing cloud-native deployments on AWS using Kubernetes, Docker and Terraform. Proven track record in optimizing SQL/NoSQL databases and automating CI/CD pipelines (Jenkins, ArgoCD) to accelerate delivery. Deeply focused on institutional security (OAuth 2.0, PCI-DSS) and enterprise observability (Prometheus, ELK, Datadog) for mission-critical trading systems.

TECHICAL SKILLS:

Programming Languages: Java, TypeScript, JavaScript
Frontend Development: React.js, Next.js, Redux Toolkit, Tailwind CSS, Material UI
Backend Development: Spring Boot, Spring MVC, Node.js, NestJS, Express.js
API Development: RESTful APIs, GraphQL
Microservices & Event Streaming: Microservices Architecture, Apache Kafka, RabbitMQ
Security & Authentication: Spring Security, OAuth 2.0, OpenID Connect, JWT
Databases & Caching: PostgreSQL, Oracle Database, MongoDB, Redis
Cloud Platforms: AWS (EC2, S3, RDS, DynamoDB, Lambda, API Gateway, CloudFront)
Infrastructure as Code: Terraform, AWS CloudFormation
Containers & Orchestration: Docker, Kubernetes, Helm
CI/CD & DevOps: Jenkins, GitHub Actions, GitLab CI/CD, ArgoCD
Testing & QA: JUnit, Mockito, TestNG, Jest, Cypress, Selenium, REST Assured, Postman
Monitoring & Observability: Prometheus, Grafana, ELK Stack, Splunk, Datadog

EDUCATION:

Bachelor of Science in Computer Science, Marymount University, Virginia, USA

PERSONAL PROJECT:

JobTrackr Pro-
Full-Stack Job Application Tracking Platform | Nov 2025 - Jan 2026

- Built and deployed a production-ready web application for managing job application pipelines with secure authentication & analytics dashboards.
- Developed RESTful APIs using Node.js/Express & MongoDB with JWT auth and user-scoped data.
- Created responsive UI with React + TypeScript and integrated charts using Recharts.
- Deployed the application on Vercel and Render with environment-based configuration and cross-origin API integration.

Shuttle Tracking Web Application

- Built a responsive web application to display shuttle routes, schedules and tracking data.
- Developed front-end components using JavaScript with a focus on clean UI/UX.

PROFESSIONAL EXPERIENCE:

State Street - NC, USA | Software Engineer | Sep 2025 – Present

- Developed a microservices-based financial platform using Java, Spring Boot, Spring MVC and Node.js, enabling scalable trade processing and portfolio analytics across internal investment operations teams.
- Built high-performance React.js + TypeScript front-end applications using Next.js, Redux Toolkit, Tailwind CSS and Material UI, delivering dynamic institutional dashboards that visualized real-time portfolio exposure, trade lifecycle and compliance alerts.
- Designed and implemented RESTful and GraphQL APIs with Spring Boot, NestJS and Express.js, enabling seamless integration between portfolio management systems, market data services and internal compliance platforms.
- Engineered an event-driven trade processing pipeline using Apache Kafka and RabbitMQ, enabling reliable asynchronous processing of over millions of institutional trade events per day
- Implemented secure authentication and authorization frameworks using Spring Security, OAuth 2.0, OpenID Connect and JWT, ensuring secure financial data access aligned with PCI-DSS and institutional security policies.
- Developed high-performance database layers using PostgreSQL, Oracle Database, MongoDB and Redis, optimizing query execution and caching strategies to reduce analytics query latency by 45%.
- Deployed cloud-native applications on AWS (EC2, S3, RDS, DynamoDB, Lambda, API Gateway, CloudFront) to support highly available and scalable financial workloads used by global institutional clients.
- Automated infrastructure provisioning using Terraform & AWS CloudFormation, standardizing environment setup across development, staging and production environments while improving deployment consistency.
- Implemented containerized microservices architecture using Docker, Kubernetes & Helm, enabling automated scaling & orchestration of trading & analytics services across multiple clusters
- Built robust CI/CD pipelines using Jenkins, GitHub Actions, GitLab CI/CD and ArgoCD, reducing production deployment time by 60% and enabling multiple weekly releases.
- Established comprehensive testing frameworks with JUnit, Mockito, Jest, Cypress, Selenium, TestNG, REST Assured and Postman, achieving 90%+ test coverage across critical financial transaction workflows.
- Implemented enterprise observability using Prometheus, Grafana, ELK Stack, Splunk and Datadog, enabling proactive monitoring and reducing production incident response time by 35% for trade processing services.

ACADEMIC PROJECT:

Marymount University | Software Engineer Intern | Jan 2026 - Present

- Developed and improved a full-stack web application for the GoVA project using Java, MySQL and IBM Cloud, which made it easier for the team to manage, search and report structured data.
- Designed and implemented efficient data structures to store and retrieve application data.
- Built a responsive Java GUI for creating, updating, searching and generating reports, improving user workflow efficiency.
- Collaborated with cross-functional team members using Git and CI/CD pipelines to debug issues, test features and improve system usability, leading to quicker issue resolution
- Applied software-engineering best practices such as version control with Git, modular design and code reviews, which increased code quality and reduced rework.

LEADERSHIP & ACTIVITIES:

Hackathon Club Member - Marymount University, Virginia, USA

- I led small team discussions during hackathons to define project scope and technical approach
- I took initiative in debugging, integrating APIs and ensuring timely project completion

Cyberethics Club Member - Marymount University, Virginia, USA

- Participated in discussions on cybersecurity ethics, data privacy & responsible use of technology
- Collaborated with peers to analyze real-world cases involving ethical hacking, AI bias & digital security
- Contributed to awareness initiatives promoting safe online practices among students
- Attended workshops & guest lectures on cybersecurity policies, compliance & emerging threats

ACHIVEMENTS:

Developed AI-Powered Resume Optimizer & Builder [LINK]